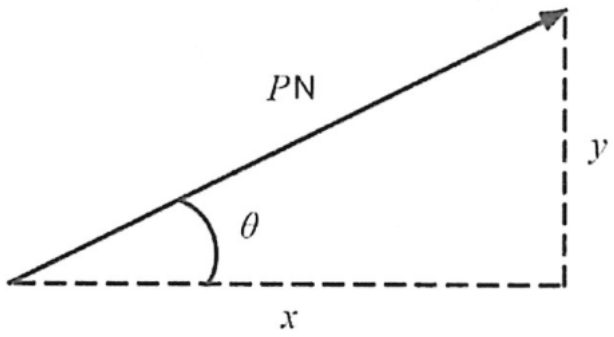
Forces

Questions 1–91

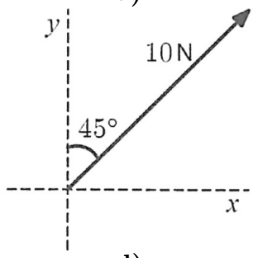
2.52 — Question 1. Find the horizontal and vertical component x and y of the force P as shown in the diagram below, for each of the given values of P and θ:



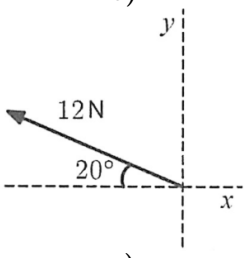
a) $P = 5$ N, $\theta = 30°$
b) $P = 6$ N, $\theta = 45°$
c) $P = 8$ N, $\theta = 40°$
d) $P = 3$ N, $\theta = 22°$

2.52 — Question 2. Find the horizontal and vertical component of each of the forces shown:

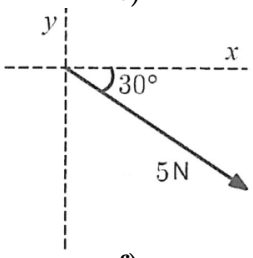
a)



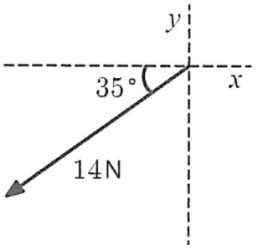
b)



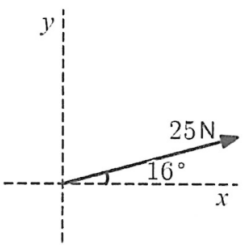
c)



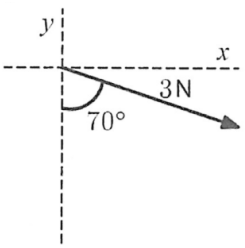
d)



e)

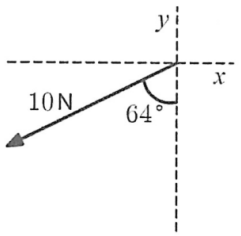


f)

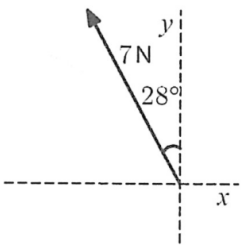


2.52 — Question 2 (continued). Find the horizontal and vertical component of each of the forces shown:

g)

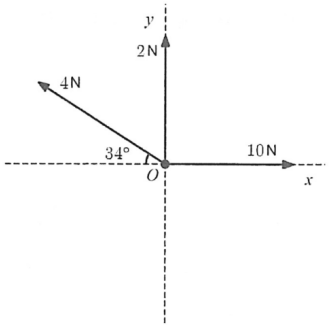


h)

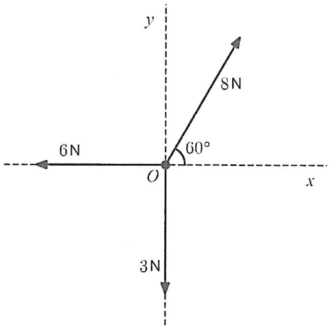


2.52 — Question 3. For each of the diagrams below, a particle is acted upon by the forces shown. Find the magnitude of the resultant force and the angle the resultant makes with Ox:

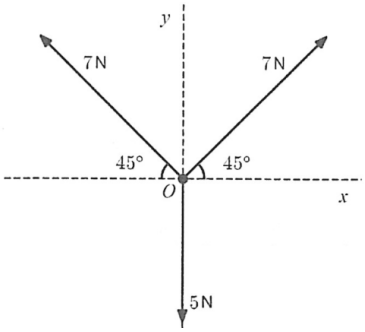
a)



b)

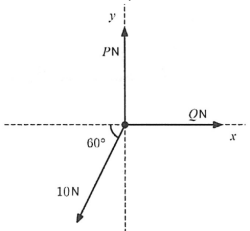


c)

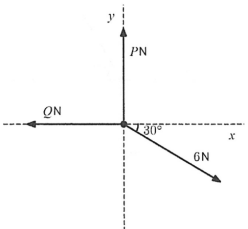


2.52 — Question 4. In each of the diagrams below, a particle is at rest, under the action of the forces shown.
In each case, by resolving in the x-direction and y-direction, find the magnitude of any unknown forces (labelled P, Q, R) and the size of any unknown angles (labelled θ):

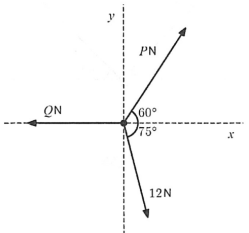
a)



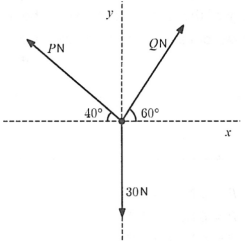
b)



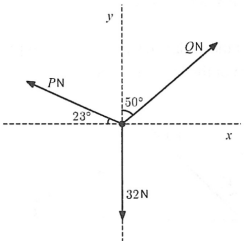
c)



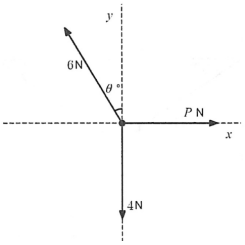
d)



e)

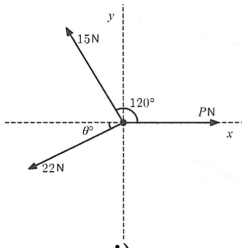


f)

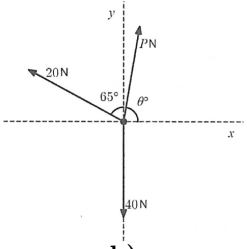


2.52 — Question 4 (continued). In each of the diagrams below, a particle is at rest, under the action of the forces shown.
In each case, by resolving in the x-direction and y-direction, find the magnitude of any unknown forces (labelled P, Q, R) and the size of any unknown angles (labelled θ):

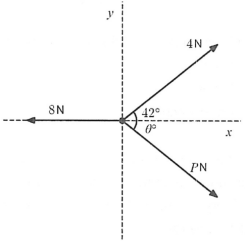
g)



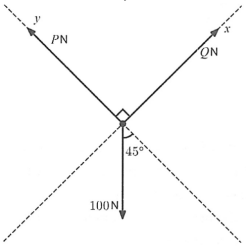
h)



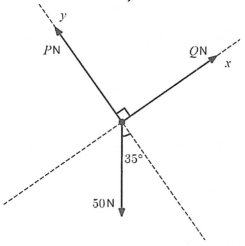
i)



j)

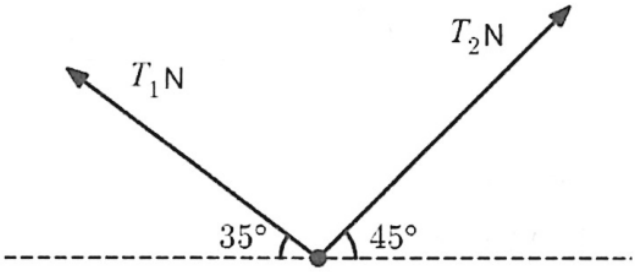


k)

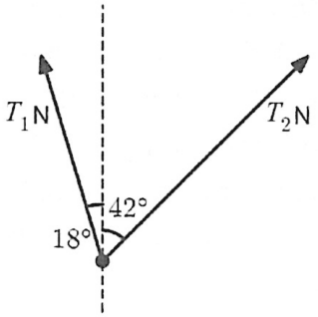


2.52 — Question 5. The following questions use $g = 9.8$ **ms**$^{-2}$ An object of mass 2 kg is held in equilibrium, suspended by two light, inextensible strings as shown in each of the diagrams. The tensions in the string are given by T_1 N and T_2 N.
In each case, find the value of T_1 and T_2

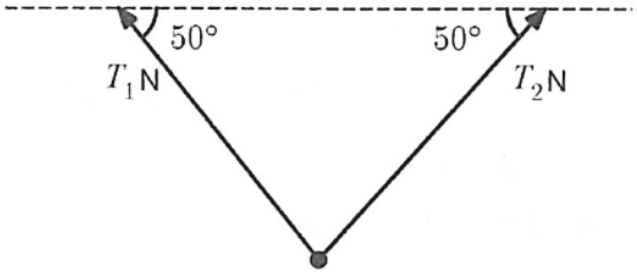
a)



b)

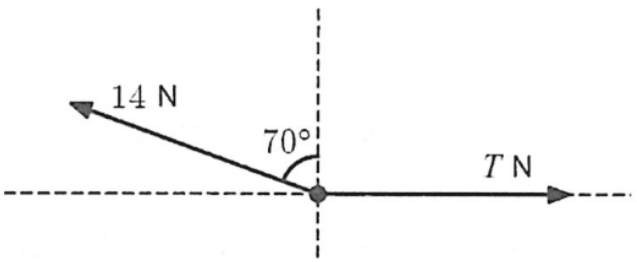


c)

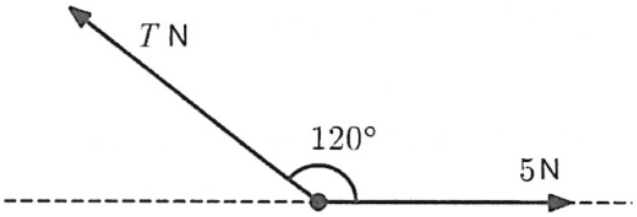


2.52 — Question 6. A particle of mass m kg is held in equilibrium, suspended by two light, inextensible strings as shown in each of the diagrams. One of the strings has tension T
In each case, find the value of T and m

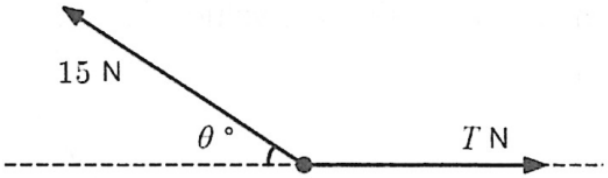
a)



b)



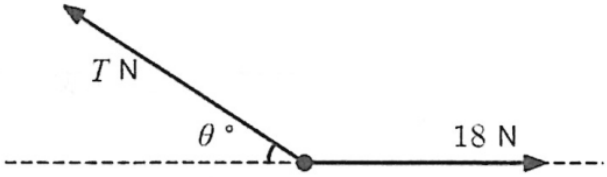
2.52 — Question 7. A particle of weight 10 N is held in equilibrium, suspended by two light, inextensible strings as shown in the following diagram. The tension in the horizontal string is T N. The second string is pulling with a tension of 15 N at $\theta°$ to the horizontal in the opposite direction to T



Find

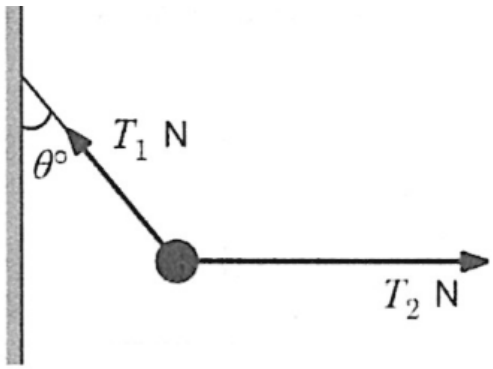
a) the value of θ
b) the value of T

2.52 — Question 8. A particle of weight 4 N is held in equilibrium, suspended by two light, inextensible strings as shown in the following diagram. The tension in the horizontal string is 18 N. The second string is pulling with a tension of T N at $\theta°$ to the horizontal in the opposite direction to the first string.



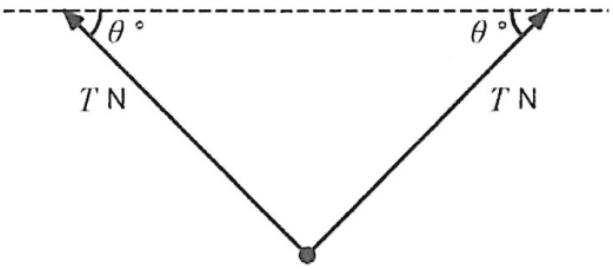
a) Find the value of θ
b) Find the value of T

2.52 — Question 9. The following questions use $g = 10$ **ms**$^{-2}$ A box of mass 4 kg is suspended by a cable, which makes an angle $\theta°$ with a vertical wall. A horizontal rope keeps the box in equilibrium.



a) Calculate the tension in the cable giving your answer in terms of θ
b) Calculate the tension in the rope giving your answer in terms of θ
c) Find the angle at which the tension in the cable and the rope is the same.
d) Find the range of angles at which the tension in the rope is less than the weight of the box.

2.52 — Question 10. The following questions use exact values of g A particle of mass m kg is held in equilibrium by two strings, each inclined at $\theta°$ to the horizontal. The tension in each string is T N.

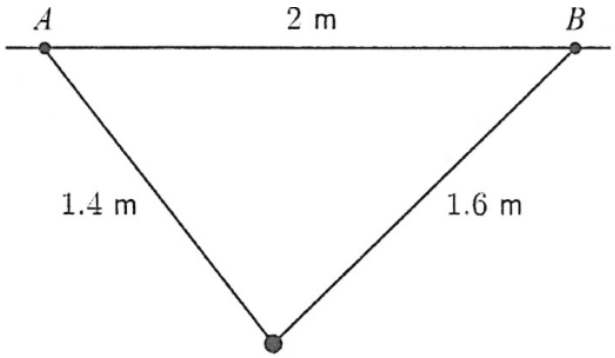


a) Show that

$$T = \frac{mg}{2\sin\theta°}$$

Find T when the particle has weight 5 N and
b) θ is 45°
c) θ is 60°
d) $\theta°$ approaches 90°

2.52 — Question 11. Two ropes are hung from a ceiling, 2 metres apart and are connected at their other ends. The length of rope A and rope B are 1.4 metres and 1.6 metres respectively.



A crate of mass m kg is attached at the end of both rope A and rope B. The system is in equilibrium. Show that

a) the acute angle α between rope A and the ceiling is given by

$$\cos\alpha = \frac{17}{28}$$

b) the acute angle β between rope B and the ceiling is given by

$$\cos\beta = \frac{23}{32}$$

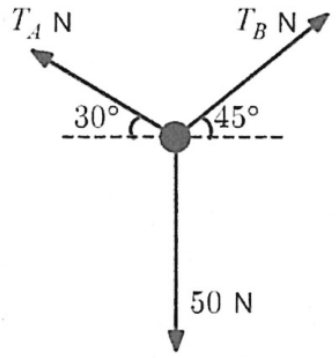
The tensions in rope A and rope B are given by T_A and T_B respectively.

c) Show that $\frac{T_B}{T_A} = \frac{136}{161}$

Given that $T_A = 322$ N,

d) find the exact value of m in terms of g

2.52 — Question 12. The following questions use Triangles of Forces and $g = 9.8$ ms^{-2} An object of weight 50 N is suspended, in equilibrium, by two light inextensible strings A and B which make angles of 30° and 45° with the horizontal as shown in the diagram below:

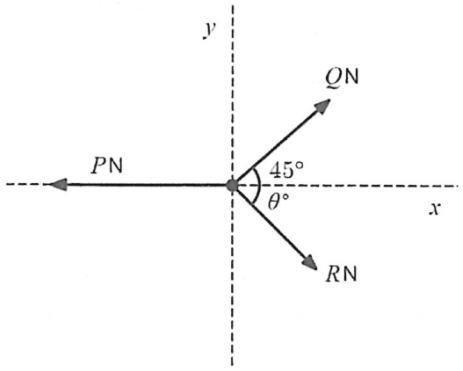


The tension in string A is given by T_A N and the tension in string B is given by T_B N.

a) Sketch the triangle of forces for these forces.

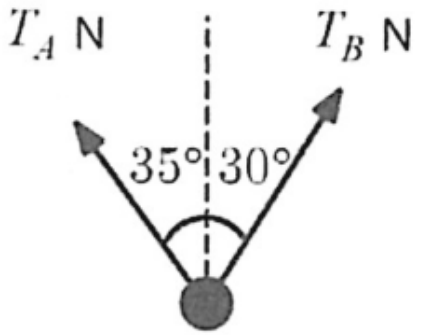
b) Hence determine the value of T_A and T_B

2.52 — Question 13. The diagram below shows a system of forces in equilibrium:



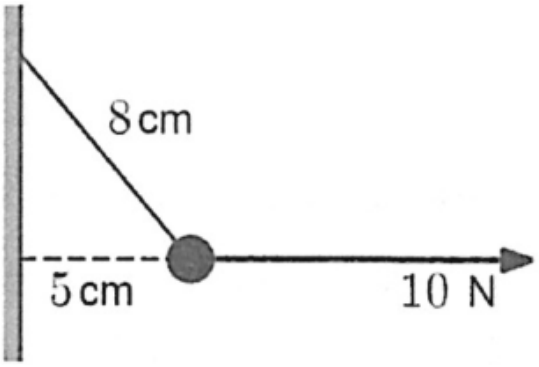
a) Sketch the triangle of forces for these forces.
b) Find the range of possible values of θ for the system to remain in equilibrium.

2.52 — Question 14. A particle of mass 2 kg is suspended by two light inextensible strings A and B which make an angles of 35° and 30° to the vertical respectively, as shown in the diagram below:



The tension in string A is given by T_A N and the tension in string B is given by T_B N.
a) Sketch the triangle of forces for the forces acting on the object.
b) Hence determine the value of T_A and T_B

2.52 — Question 15. An object of weight w N is attached to one end of a light inextensible string of length 8 cm. The other end is attached to a vertical wall at the point A. A horizontal force of magnitude 10 N is applied to the object. The particle is in equilibrium with the string taut, where it lies 5 cm from the wall.



a) Sketch a triangle of forces for the forces acting on the object.

Hence find

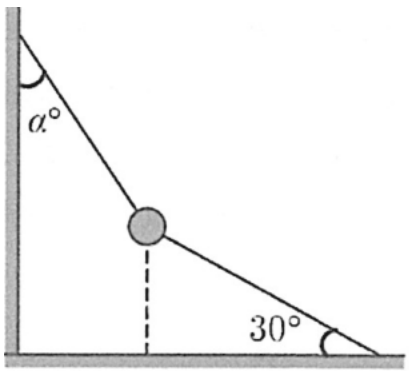
b) the tension in the string.

c) the value of w

d) the mass of the object.

2.52 — Question 16. The following questions use exact values of g

Prerequisite knowledge for Q16: Compound Angle Formulae A small box B of mass m kg is held in equilibrium by two light strings, one attached to a vertical wall which has a tension of T_W N and one to the horizontal floor which has a tension of T_F N. They make an angle of $\alpha°$ with the wall and 30° with the floor respectively.



a) Sketch a triangle of forces for the forces acting on the box.

Hence or otherwise, given $\alpha = 30°$, show that

b) $T_W = \sqrt{3}mg$

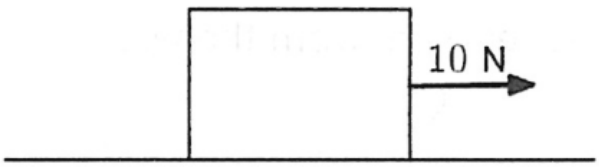
c) $T_F = mg$

2.52 — Question 16 (continued). Angle α is increased and the angle of the string with the floor becomes 45°. The system remains in equilibrium.

d) Show that

$$T_W = \frac{mg}{\cos \alpha^\circ - \sin \alpha^\circ}$$

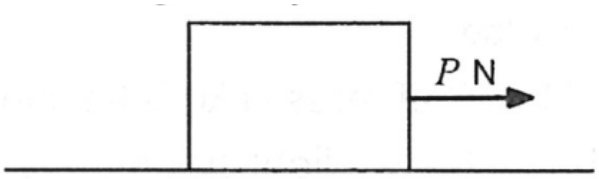
2.53 — Question 1. A particle of weight w N lies on a rough horizontal surface. A horizontal force 10 N is applied to the particle. The coefficient of friction between the particle and the surface is given by μ



The frictional force F N is the only resistance force, and the particle is at the point of moving. Find the value of F given

a) $w = 50, \mu = 0.2$
b) $w = 20, \mu = 0.45$
c) $w = 15, \mu = 0.85$
d) $w = 3, \mu = 1.1$

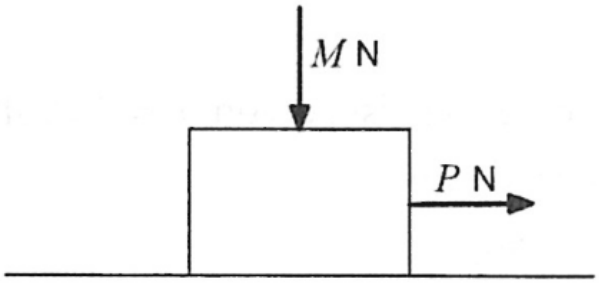
2.53 — Question 2. A crate of weight w N lies on a rough horizontal surface. A horizontal force P N is applied to the crate. The frictional force F N is the only resistance against the crate. The coefficient of friction between the crate and the surface is given by μ



Given the crate is at the point of moving and

a) $w = 12, \mu = 0.5$, find the value of P
b) $w = 12, \mu = 0.9$, find the value of P
c) $P = 20, \mu = 0.4$, find the value of w
d) $P = 20, \mu = 0.8$, find the value of w
e) $w = 50, P = 40$, find the value of μ
f) $w = 300, P = 12$, find the value of μ
g) $w = 12, P = 15$, find the value of μ

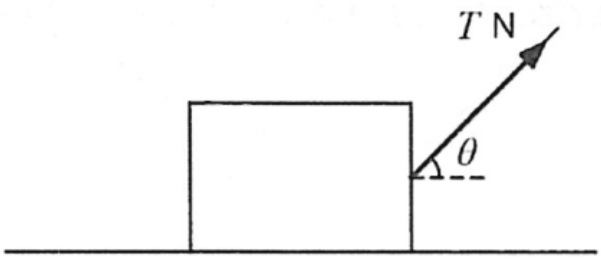
2.53 — Question 2 (continued). The crate has objects added to it with weight M N as shown in the diagram below. The other forces on the crate remain the same.



Given the crate is at the point of moving and

h) $w = 12, \mu = 0.7, M = 3$, find the value of P
i) $P = 8, \mu = 0.8, M = 2$, find the value of w
j) $w = 60, P = 10, M = 20$ find the value of μ

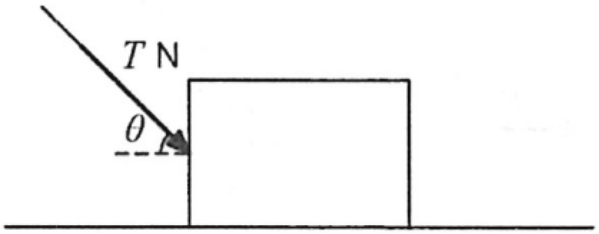
2.53 — Question 3. A box of weight w N lies on a rough horizontal surface. It is being pulled by a horizontal force T N at an angle $\theta°$ to the horizontal. The frictional force F N is the only resistance against the box. The coefficient of friction between the box and the surface is given by μ



Given the box is at the point of moving and

a) $w = 20, T = 15, \theta = 30, \mu = 0.5$, find the value of F
b) $w = 8, \theta = 45, \mu = 0.2$, find the value of T
c) $T = 10, \theta = 60, \mu = 0.1$, find the value of w
d) $w = 15, T = 4, \theta = 30$, find the value of μ

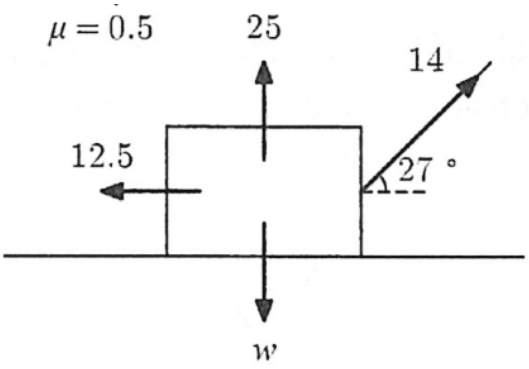
2.53 — Question 4. A box of weight w N lies on a rough horizontal surface. It is being pushed by a force T N at an angle $\theta°$ to the horizontal. The frictional force F N is the only resistance against the box. The coefficient of friction between the box and the surface is given by μ



Given the box is at the point of moving and

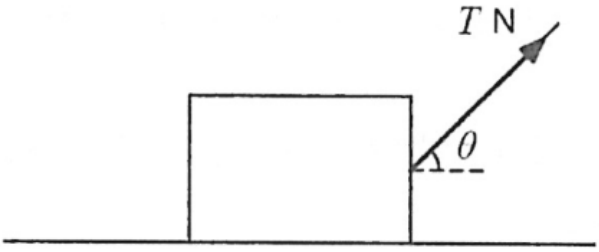
a) $w = 2, T = 4, \theta = 30, \mu = \frac{1}{2}\sqrt{3}$, find the value of F
b) $w = 4, \theta = 45, \mu = 0.3$, find the value of T
c) $T = 12, \theta = 60, \mu = 0.2$, find the value of w
d) $w = 20, T = 5, \theta = 30$, find the value of μ

2.53 — Question 5. Will has drawn a correct force diagram for a particle in equilibrium, and some workings:



$R(\uparrow)$: $25 + 14\sin 27° - w = 0 \therefore$ mass is 31.4 kg
Explain why his claim that the mass of the particle is 31.4 kg is incorrect.

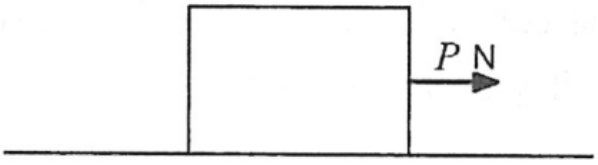
2.53 — Question 6. The following questions use $g = 9.8$ ms^{-2} A box of mass m kg lies on a rough horizontal surface. It is being pulled by a force T N at an angle $\theta°$ to the horizontal. The frictional force F N is the only resistance against the box. The coefficient of friction between the box and the surface is given by μ



Given the box is at the point of moving and

a) $m = 5, \theta = 10, \mu = 0.5$, find the value of T
b) $T = 20, \theta = 50, \mu = 0.1$, find the value of m
c) $m = 2, T = 5, \theta = 70$, find the value of μ

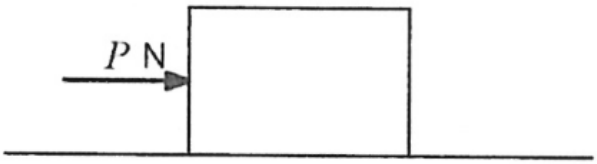
2.53 — Question 7. The coefficient of friction between a particle of mass 2 kg, on a rough horizontal surface is 0.4. A horizontal force of P N acts on the particle.



In each case, identify if the particle is moving under the given value of P and whether, when an additional mass of 100 g is added to the particle, it continues to move or remain stationary.

a) $P = 8.4$
b) $P = 8$
c) $P = 7.6$

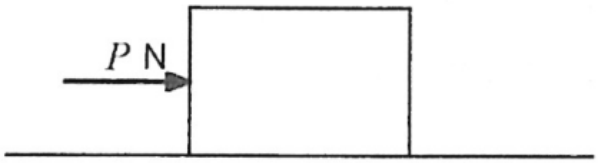
2.53 — Question 8. A crate of mass m kg lies on a rough floor. The coefficient of friction between the crate and the floor is 0.3. The crate is pushed by a horizontal force P N and it does not move.



For each of the following, find the greatest possible values of P

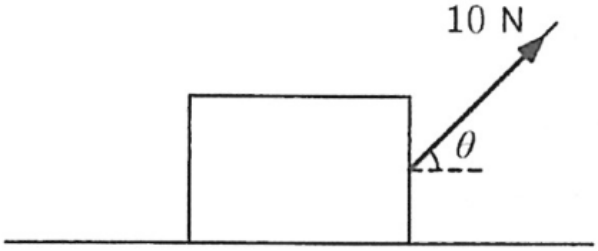
a) $m = 5$ kg
b) $m = 10$ kg
c) $m = 20$ kg

2.53 — Question 9. The following questions use exact values of g A box of mass m kg lies on a rough floor. The coefficient of friction between the crate and the floor is μ. The crate is pushed by a horizontal force P N and it does not move.



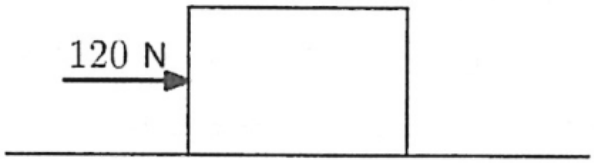
Show that $P \leq \mu mg$

2.53 — Question 10. The following questions use $g = 10\ \text{ms}^{-2}$ A block of mass 15 kg lies on a rough horizontal surface. The block is being pulled by a force of 10 N at an angle $\theta°$ to the horizontal. The coefficient of friction between the block and the mass is 0.1



Given the block is moving, show that $10\cos\theta + \sin\theta > 15$

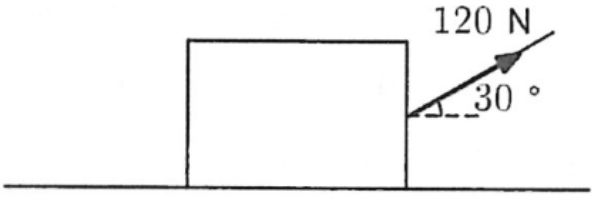
2.53 — Question 11. A block of mass 15 kg lies on a rough horizontal surface. The coefficient of friction between the block and the mass is 0.85



The block is pushed by a horizontal force of 120 N.

a) Does the block move, and if so, what is it acceleration?

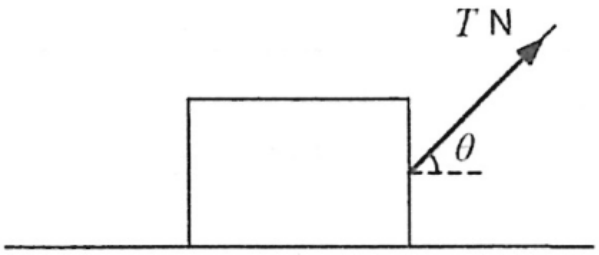
The pushing force is removed, and instead a pulling force of 120 N at angle of 30° to the horizontal is applied.



b) Find the acceleration of the block.

c) Find the maximum extra mass m kg which may be added to the block, for it still to be moving.

2.53 — Question 12. The following questions use exact values of g A box of mass m kg lies on a rough horizontal surface. It is being pulled by a horizontal force T N at an angle $\theta°$ to the horizontal. The coefficient of friction between the box and the surface is μ. The frictional force F N is the only resistance against the box.

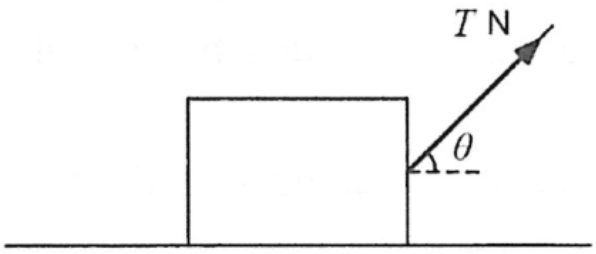


Show that $T \geq \frac{\mu m g}{\cos\theta + \mu \sin\theta}$

2.53 — Question 13. For each of the following, find the exact values of $\sin\theta$ and $\cos\theta$:

a) $\tan\theta = \frac{3}{4}$
b) $\tan\theta = \frac{5}{12}$
c) $\tan\theta = \frac{7}{16}$
d) $\tan\theta = \frac{10}{21}$

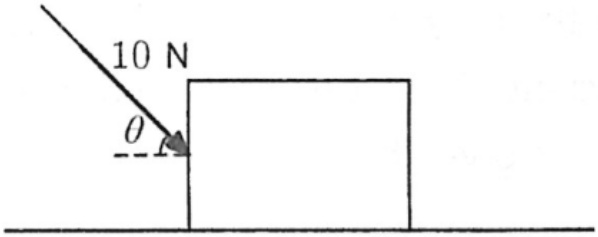
2.53 — Question 14. The following questions use $g = 9.8$ ms^{-2} A box of mass m kg lies on a rough horizontal surface. It is being pulled by a force T N at an angle $\theta°$ to the horizontal where $\tan\theta = \frac{3}{4}$.
The coefficient of friction between the box and the surface is μ. The frictional force F N is the only resistance against the box.



Given the box is moving and the acceleration of the box is a ms^{-2}, and

a) $m = 10, T = 20, \mu = 0.1$, find the value of a
b) $m = 3, T = 20, \mu = 0.2$, find the value of a
c) $m = 5, T = 15, a = 1.5$, find the value of μ
d) $T = 16, \mu = 0.25, a = 3$, find the value of m
e) in general, show that $a = (\frac{4}{5m} + \frac{3}{5m}\mu)T - 9.8\mu$

2.53 — Question 15. A box of mass 4 kg lies on a rough horizontal surface. It is being pushed by a force 10 N at an angle $\theta°$ to the horizontal where $\tan\theta = \frac{5}{12}$
The coefficient of friction between the box and the surface is 0.2.



a) Find the acceleration of the box.

The box has additional mass of m kg added to it, given

b) $m = 0.1$ find the value of a

c) $m = 0.5$ find the value of a

2.53 — Question 16. A trolley of mass 8 kg and is being pushed with a horizontal force of P N along a rough horizontal floor. It is at the point of moving.
In doubling the horizontal pushing force, the trolley moves with acceleration 0.25 ms^{-2}. Find

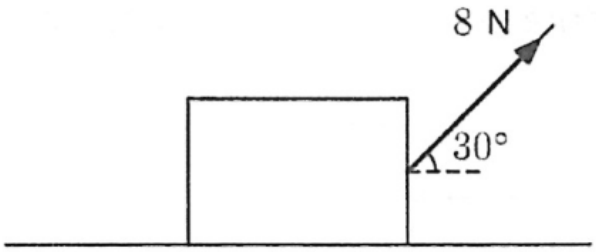
a) the value of P

b) the coefficient of friction between the trolley and the floor.

The trolley is loaded with an extra m kg of products. The trolley is pushed at $2P$ N.

c) Find the maximum value of m such that the trolley is moving or on the point of moving.

2.53 — Question 17. A sledge of mass 1.2 kg lies on a rough horizontal ground and is being pulled by a force 8 N at an angle 30° to the horizontal, having initially started at rest. The coefficient of friction between the sledge and the ground is 0.2

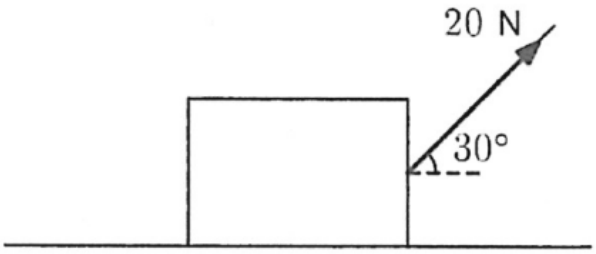


a) Find the acceleration of the sledge.

After 5 seconds, the rope snaps. The sledge continues to move until it comes to a stop.

b) Find the distance the sledge travels after the rope breaks.

2.53 — Question 18. The following questions use $g = 9.81\ \text{ms}^{-2}$ A crate of mass 5 kg lies on a rough horizontal surface. It is being pulled by a force 20 N at an angle 30° to the horizontal. The coefficient of friction between the crate and the surface is 0.4

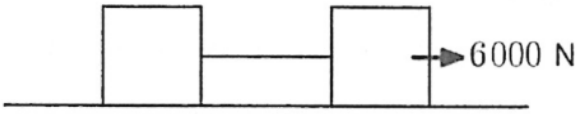


a) Find the acceleration of the crate.

The 20 N force needs to be replaced by a horizontal pushing force P N such that it moves at the same acceleration.

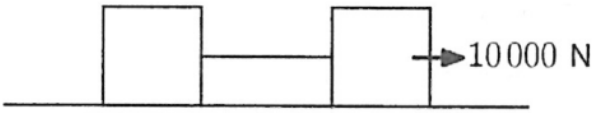
b) Find the value of P

2.53 — Question 19. A car of mass 2000 kg is towing a trailer of mass 500 kg along a straight flat road. The driving force produced by the engine is 6000 N. The coefficient of friction between the road and the vehicles is 0.1



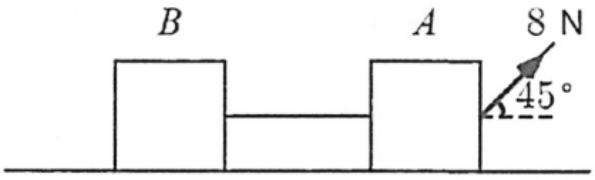
a) Draw a force diagram.
b) Find the acceleration of the car.
c) Find the tension in the tow bar.
d) State any modelling assumptions you have made.

2.53 — Question 20. A van of mass 3500 kg is towing a trailer of mass 750 kg along a straight flat road. The driving force produced by the engine is 10000 N. The tow bar is modelled as both light and inextensible.



The van is accelerating at 1.5 ms^{-2}.
The coefficient of friction between both the van and the road, and the trailer and the road is μ
a) Find the value of μ
b) Find the tension in the tow bar.
c) Explain how the modelling assumptions that the string is light and inextensible have been used.

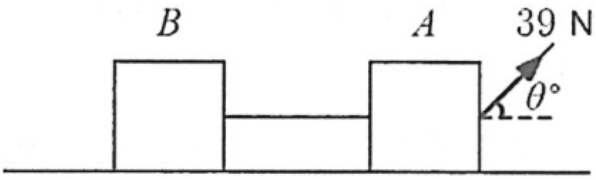
2.53 — Question 21. The following questions use $g = 9.8$ ms^{-2} A toy train is made up of an engine A of mass 0.8 kg connected by a horizontal, light, inextensible string to truck B of mass 0.3 kg.
The engine is being pulled by a light inextensible string with tension 8 N which is at an angle of 45° to the horizontal.



The coefficient of friction between both the engine and the truck and the floor is 0.95

a) Draw a force diagram.
b) Find the friction force acting on the engine.
c) Find the friction force acting on the truck.
d) Find the acceleration of the engine.
e) Find the tension in the string connecting the engine and the truck.
f) State a modelling assumption you have used.

2.53 — Question 22. A particle A of mass 10 kg and a particle B of mass 5 kg are on a horizontal rough surface and are connected by a light, inextensible string which remains taut.
Particle A is being pulled by a light, inextensible string, at an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$. The tension in this string is 39 N.



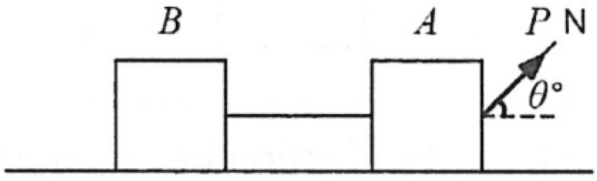
The coefficient of friction between the particles and the surface is μ.
The deceleration of the particles is 3.5 ms^{-2}.

a) Draw a force diagram.
b) Find the coefficient of friction between the particles and the surface.
c) Find the tension in the string connecting the two particles.
d) State a modelling assumption that you have used.

It takes 10 seconds before coming to rest.

e) Find the distance they have travelled.

2.53 — Question 23. The following questions use $g = 10$ ms^{-2} A particle A of mass 8 kg and a particle B of mass 3 kg are on a horizontal rough surface and are connected by a light, horizontal, inextensible string which remains taut.
Particle A is being pulled by a light, inextensible string, at an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$. The tension in this string is P N.



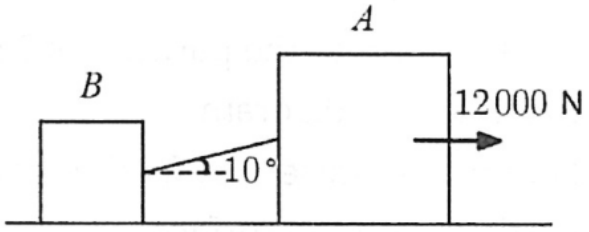
The coefficient of friction between the particles and the surface is μ.
The system is initially at rest, then the acceleration of the particles is 2.5 ms^{-2}.

a) Draw a force diagram.

It is given that the friction acting against the motion of particle B is 6 N. Find

b) the value of μ
c) the tension in the string connecting particle A and the particle B.
d) the value of P.
e) the friction acting against particle A.
f) How long it takes for the particles to travel 20 metres.

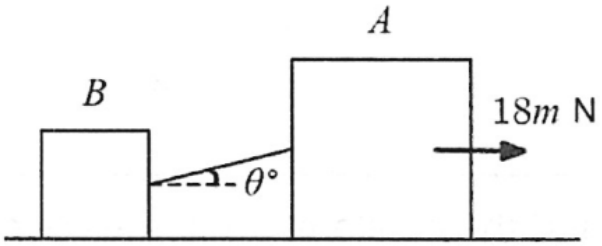
2.53 — Question 24. A van of mass 4000 kg is towing a trailer B of mass 1000 kg along a straight flat road. The driving force produced by the engine is 12000 N.
The tow bar is angled upwards from B to A with angle 10° to the horizontal. The tow bar may be modelled as a rod.



The coefficient of friction between the van and trailer and the surface is μ
The van is accelerating at 0.5 ms^{-2}.

a) Draw a force diagram.
b) Find the value of μ
c) Find the tension in the tow bar.

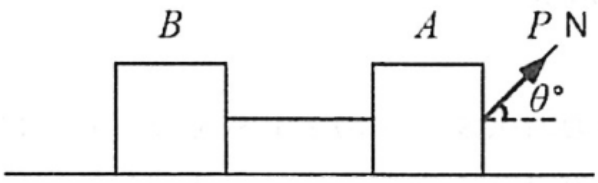
2.53 — Question 25. A particle A of mass m kg and a particle B of mass $3m$ kg are on a horizontal rough surface and are connected by a light, horizontal, inextensible string which remains taut. Particle A is pulled forward by a force of $18m$ N.
The tow bar is angled upwards from B to A with angle $\theta°$ to the horizontal. The tow bar may be modelled as a rod.



The coefficient of friction between the van and trailer and the surface is μ
The van is accelerating at 2 ms^{-2}.

a) Show that $\mu = 0.25$
b) Show that $T = \frac{54m}{\sin\theta + 4\cos\theta}$

2.53 — Question 26. The following questions use exact values of g A particle A of mass $4m$ kg and a particle B of mass m kg are on a horizontal rough surface and are connected by a light, inextensible string. Particle A is being pulled by a force of P N, at an angle of $\theta°$ to the horizontal where $\tan\theta = \frac{3}{4}$. The coefficient of friction between the particles and the surface is μ.

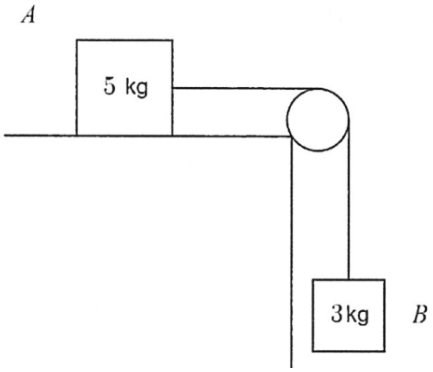


a) Draw a force diagram.

Given that $P = 25$ and the total resistance force is 10 N.

b) Show that $\mu = \frac{2}{mg - 3}$
c) Show that the acceleration of the system is given by $a = \frac{2}{m}$ ms^{-2}
d) Find the tension in the string between the two particles in terms of m

2.54 — Question 1. The following questions use $g = 9.8$ **ms**$^{-2}$ Two particles, A and B of mass 5 kg and 3 kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string.

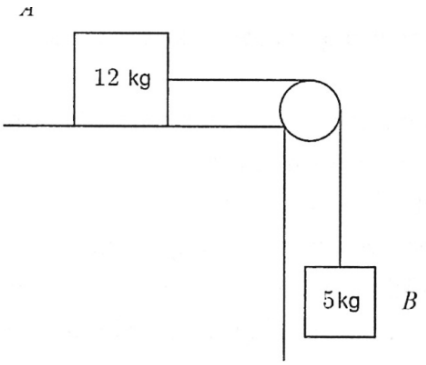


The coefficient of friction between A and the table is 0.25.

a) Find the acceleration of the system.

b) Find the tension in the string.

2.54 — Question 2. Two particles, A and B of mass 12 kg and 5 kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string.



The two particles are accelerating at 0.02 ms^{-2}. The system is released from rest.
Find

a) the tension in the string.

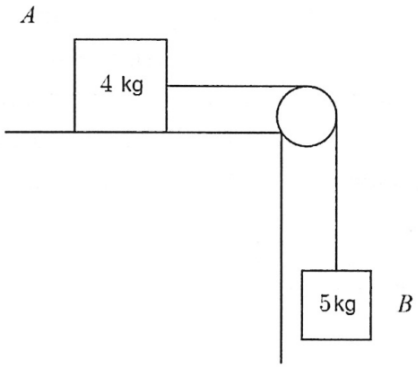
b) the coefficient of friction between A and the table.

Particle B is 0.5 m from the ground. Assuming that particle A does not reach the pulley,

c) find the time it takes for B to reach the ground.

d) state the distance A travels in this time.

2.54 — Question 3. Two particles, A and B of mass 4 kg and 5 kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string.



The coefficient of friction between A and the table is 0.4

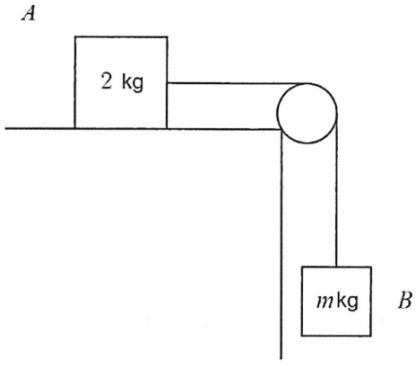
a) Find the acceleration of the system.

b) Find the tension in the string.

The system is released from rest. It takes 1.05 seconds for Particle B to reach the ground and does not reach the pulley.

c) Determine the initial height of particle B above the ground.

2.54 — Question 4. The following questions use exact values of g Two particles, A and B of mass 2 kg and m kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string.

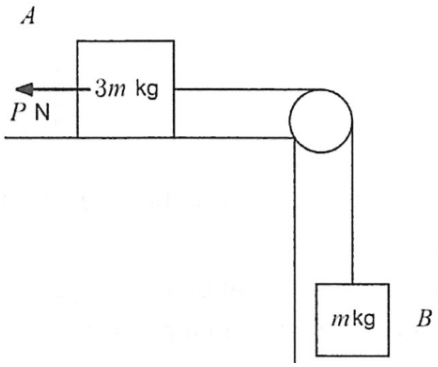


The coefficient of friction between A and the table is 0.3

a) Find the acceleration of the system in terms of m and g

b) Find the minimum value of m such that Particle B is moving.

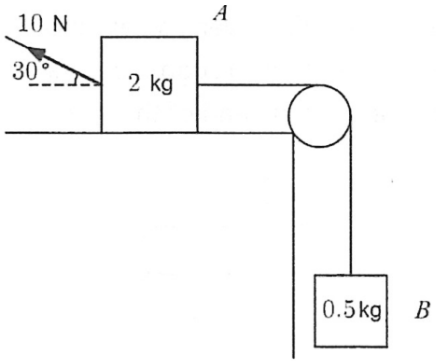
2.54 — Question 5. The following questions use $g = 9.8$ **ms**$^{-2}$ Two particles A and B of mass $3m$ kg and m kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. The coefficient of friction between A and the table is μ. Particle B hangs freely with the string taut. Particle A is pulled away from the pulley with horizontal force P N with acceleration a ms^{-2}.



Show that

$$P = (29.4\mu + 4a + 9.8)m$$

2.54 — Question 6. Two particles, A and B of mass 2 kg and 0.5 kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string. Particle A is being pulled by a force of 10 N at an angle 30° to the horizontal away from the pulley.



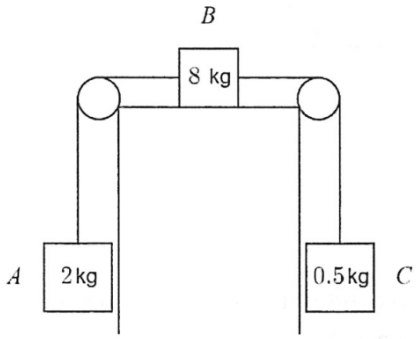
The coefficient of friction between A and the table is 0.2. Find

a) the frictional force acting on particle A
b) the acceleration of the system.
c) the tension in the string.

Particle B is swapped for particle C of mass m kg.

d) Find the maximum value of m such that C accelerate upwards.

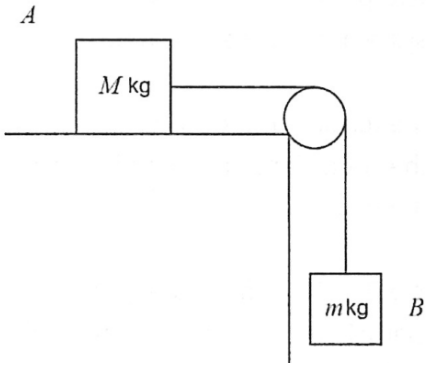
2.54 — Question 7. Three boxes, A, B and C have masses 2 kg, 8 kg and 0.5 kg respectively. Box A is hanging freely from a string, which passes over a small, fixed, smooth pulley and connects to box B. Box B is on a rough horizontal table. Box C is hanging freely from a string, which passes over a small, fixed, smooth pulley and connects to the other side of box B.



The coefficient of friction between box B and the table is 0.1. The system is moving.

a) Find the tension in the strings.
b) Find the acceleration of the system and describe the motion of the boxes.
c) State the modelling assumptions you have used.

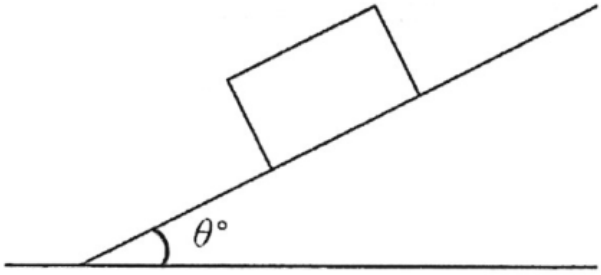
2.54 — Question 8. The following questions use exact values of g Two particles A and B of mass M kg and m kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. The coefficient of friction between A and the table is μ. Particle B hangs freely with the string taut.



The system is moving. Show that

$$m \geq \mu M$$

2.55 — Question 1. The following question use $g = 9.8$ ms^{-2} A box of mass 5 kg rests on a rough slope which is inclined with an angle of $\theta°$.



The box is at the point of slipping.

a) Draw a force diagram.

Find the coefficient of friction between the box and the slope in each case when

b) $\theta = 30$

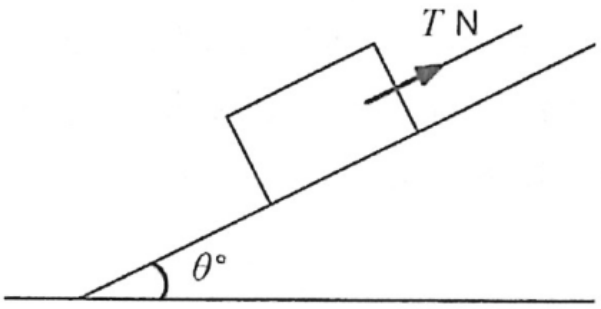
c) $\theta = 45$

d) $\theta = 55$

The box is replaced with a different box of mass m kg.

e) Show that $\mu = \tan\theta$

2.55 — Question 2. A box of mass m kg rests on a rough slope which is inclined with an angle of $\theta°$. A pulling force T N acts upwards and parallel to the slope. A frictional force F N acts parallel to the slope.



The box is at the point of slipping up the slope.

a) Draw a force diagram.

Given

b) $m = 4, \mu = 0.8, \theta = 30$, find the value of F and the value of T

c) $m = 5, \mu = 0.6, \theta = 60$, find the value of F and the value of T

d) $T = 10, \theta = 20, F = 5$, find the value of m

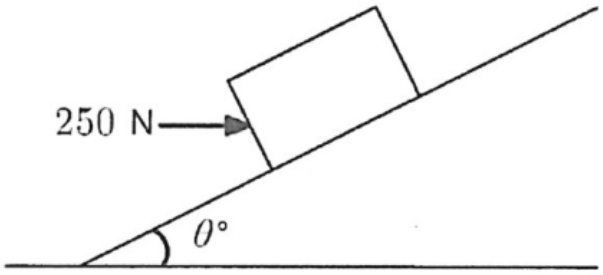
e) $T = 12, m = 2, F = 8$, find the value of θ

f) $T = 10, \mu = 0.6, \theta = 45$, find the value of m

Now using exact values of g

g) show that $T = mg(\mu\cos\theta + \sin\theta)$

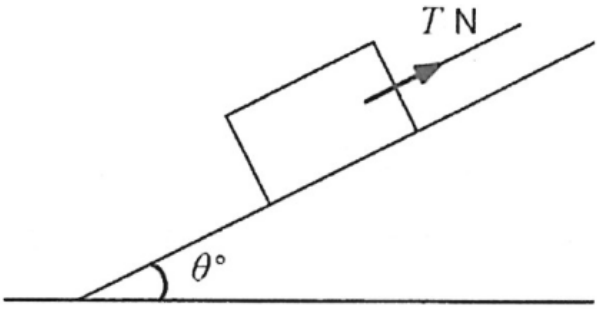
2.55 — Question 3. A particle has mass 20 kg and is held at rest on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$



It is being held at rest by a horizontal pushing force 250 N.

a) Draw a force diagram.
b) Show that $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$
c) Find the normal reaction force on the box.
d) Find the frictional force acting on the box.
e) Find the coefficient of friction acting between the box and the slope.

2.55 — Question 4. A box of weight 100 N rests on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$



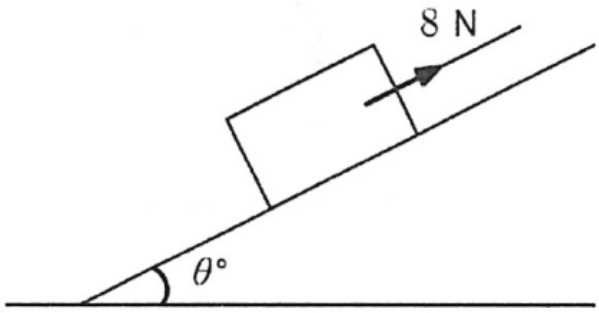
A pulling force T N acts upwards and parallel to the slope. The box is at the point of slipping down the slope.

a) Draw a force diagram.
b) Show that $\sin\theta = \frac{7}{25}$ and $\cos\theta = \frac{24}{25}$

The coefficient of friction between the box and the slope is 0.25

c) Find the value of T

2.55 — Question 5. The following question use $g = 10\ \text{ms}^{-2}$ A box of mass 5 kg rests on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{2}{5}$



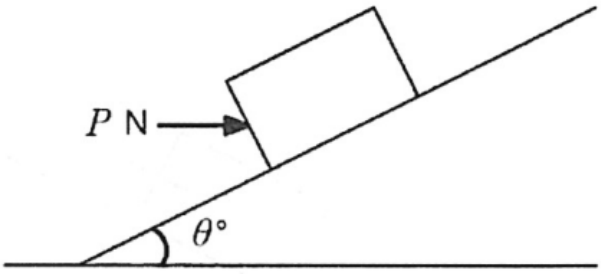
A pulling force 8 N acts upwards and parallel to the slope. The box is at the point of slipping down the slope.

a) Draw a force diagram.

b) Show that $\sin\theta = \frac{2}{\sqrt{29}}$ and $\cos\theta = \frac{5}{\sqrt{29}}$

c) Find the value of the coefficient of friction between the box and the slope.

2.55 — Question 6. A particle has mass m kg and is held at rest on a rough slope which is inclined with an angle of $\theta°$.



It is being held at rest by a horizontal pushing force P N.

a) Draw a force diagram.

Given that $m = 2, \mu = 0.4, \theta = 30$, find

b) the normal reaction force on the box in terms of P

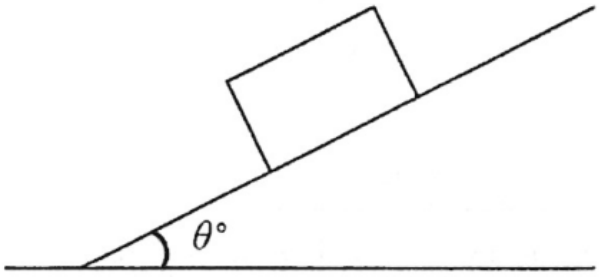
c) the frictional force in terms of P

d) the value of P

2.55 — Question 7. A sledge is released from rest on a smooth slope which is at angle 10° to the horizontal. The mass of the sledge is m kg.

a) Draw a force diagram.
b) Find the acceleration of the sledge.
c) Find the speed once it has travelled 5 metres.

2.55 — Question 8. A box of mass m kg is held at rest before being released. It lies on a rough slope which is inclined with an angle of $\theta°$. A frictional force F N acts parallel to the slope. The coefficient of friction between the box and the slope is μ



The acceleration of the box is given by a ms^{-2}.

a) Draw a force diagram.

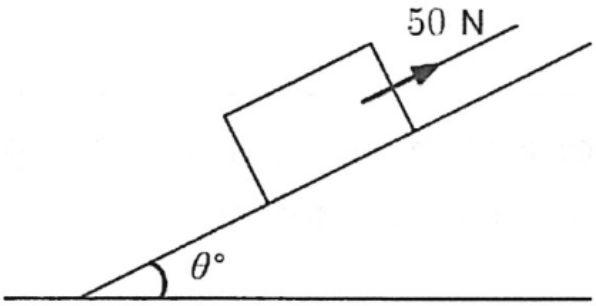
Given

b) $\theta = 60, m = 4, \mu = 0.2$, find the value of a and state its direction.
c) $\theta = 45, m = 12, a = 1.5$, find the value of μ

Now, using exact values of g,

d) show that the acceleration down the slope is given by $a = g(\sin\theta - \mu\cos\theta)$
e) State the acceleration down the slope if it were modelled as smooth.

2.55 — Question 9. A box has weight 39 N, initially at rest and on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$. It is being pulled up the slope by a rope which may be modelled as a light, inextensible string which is parallel to the slope with the tension 50 N.



The coefficient of friction between the box and the slope is given by 0.5. The acceleration of the system is a ms^{-2}.

a) Draw a force diagram.

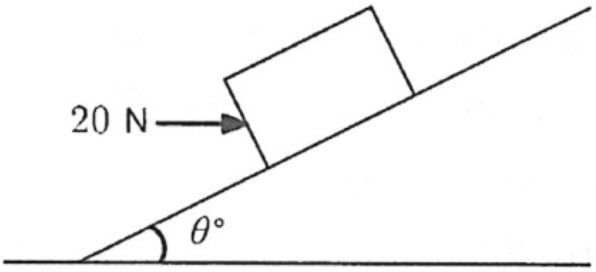
b) Show that $a = \frac{170}{39}$

The box is pulled up the slope for 1.5 seconds.

c) Find the distance travelled by the box in this time.

d) Find the speed it is moving at 1.5 seconds.

2.55 — Question 10. The following questions use $g = 9.8$ ms^{-2} A particle has mass m kg and is held at rest on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{7}$. It is being pushed up the slope by a horizontal force of 20 N which is horizontal.



The coefficient of friction between the box and the slope is given by μ. The acceleration of the system is a ms^{-2}.

a) Draw a force diagram.

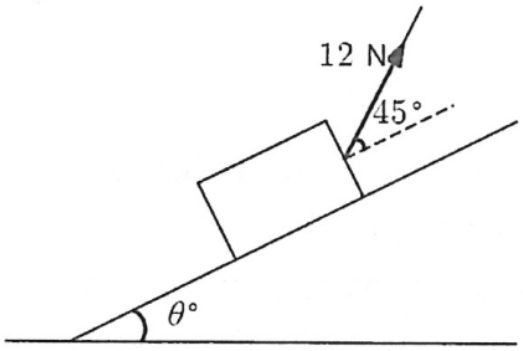
b) Find the exact values of $\cos\theta$ and $\sin\theta$

It is given that $m = 10$ and $\mu = 0.25$

c) Find the value of a

d) Describe the type and direction of the motion.

2.55 — Question 11. A particle has mass m kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$
It is being pulled up the slope by a light, inextensible string which is at angle 45° to the slope with the tension 12 N.



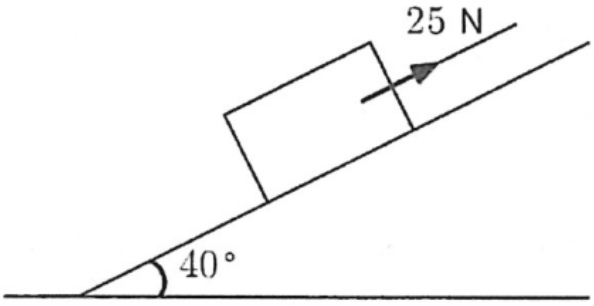
The coefficient of friction between the box and the slope is given by μ. The acceleration of the particle up the slope is 1.2 ms^{-2}.

a) Draw a force diagram.
b) Find the exact values of $\cos\theta$ and $\sin\theta$

Given

c) $m = 2$, find the value of μ
d) $\mu = 0.1$, find the value of m

2.55 — Question 12. A crate has mass 6 kg on a rough slope which is inclined with an angle of 40°. It is being pulled up the slope by a rope with the tension 25 N which is parallel to the slope. The rope may be modelled as a light, inextensible string.



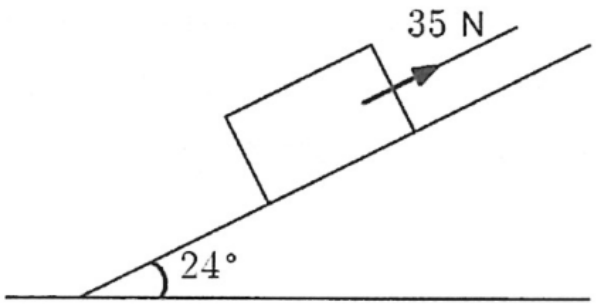
The coefficient of friction between the crate and the slope is 0.45.

a) Find the acceleration of the crate.
b) Describe the motion of the crate.

After coming to a stop, the rope is removed, and the crate slides down the slope.

c) Find the value of the acceleration down the slope.

2.55 — Question 13. A crate has mass 4 kg, initially at rest and on a rough slope which is inclined with an angle of 24°. It is being pulled up the slope by a rope with the tension 35 N which is parallel to the slope. The rope may be modelled as a light, inextensible string.



The acceleration of the system is 0.5 ms^{-2}.

a) Show that the coefficient of friction between the box and the crate is approximately 0.48

b) Find how long it takes for the crate to move 2 metres up the slope.

Upon travelling 2 metres, the end of the rope which is not attached to the crate is instantaneously adjusted upwards so that it creates an angle of 20° with the slope but remains at 35 N.

c) Find how long it takes for the crate to move a further 2 metres up the slope.

2.55 — Question 14. A toy car is placed at the end of a straight flat track of length 5 metres, held at an incline of $\theta°$ where $\tan\theta = \frac{3}{4}$. The toy car is released from rest and takes 2 seconds to reach the bottom of the track. The coefficient of friction between the car and the track is μ

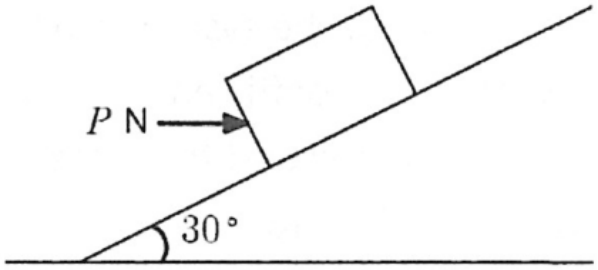
a) Find the acceleration of the toy car

b) Find the value of μ

2.55 — Question 15. A particle of mass m kg is projected up a rough slope which has an incline of $\theta°$ where $\tan\theta = \frac{12}{5}$
The coefficient of friction between the particle and the slope is 0.8. The particle initially has speed 20 ms^{-1}.
Find the time it takes for the particle to return to its original position.

2.55 — Question 16. A particle of mass m kg is projected up a rough slope which has an incline of $\theta°$ where $\tan\theta = \frac{3}{4}$.
It starts with speed 15 ms^{-1} and takes 3 s to return to its start position.
Find the coefficient of friction between the particle and the slope.

2.55 — Question 17. A block has weight 20 N and is on a rough slope which is inclined with an angle of 30°. It is being pushed up the slope by a horizontal force of P N.



The normal reaction on the block by the slope is $15\sqrt{3}$ N. The block is decelerating up the slope at 2.45 ms^{-2}.

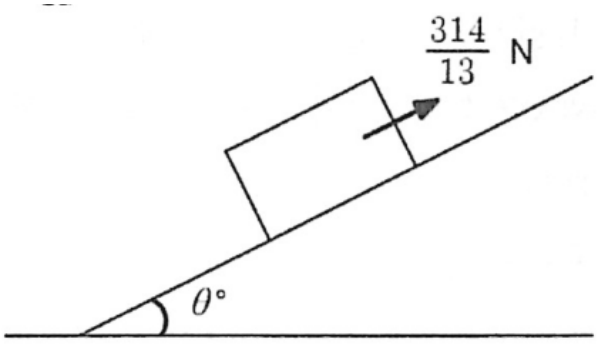
a) Find the friction force acting against the particle.

The horizontal force is removed.

b) Find the acceleration of the block.

c) Suggest improvements to the model in order to make the model more realistic.

2.55 — Question 18. A particle has weight 49 N is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$



The particle is pulled up the slope by a force parallel to the slope of $\frac{314}{13}$ N. The particle has deceleration 1.2 ms^{-2}.

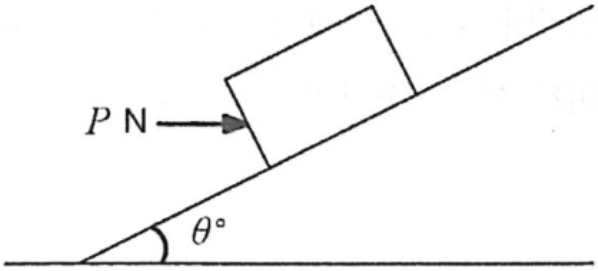
a) Find the coefficient of friction between the slope and the particle.

b) Given it takes 4 seconds to come to a stop, find the distance it travels up the slope.

The pulling force is then removed.

c) Find how long it takes for the particle to return to its original position from the point at which it came to a stop.

2.55 — Question 19. The following questions use exact values of g A particle has mass m kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{4}{3}$. It is being pushed up the slope by a horizontal force of P N.



The coefficient of friction between the particle and the slope is 0.6

a) Find the values of P such that the particle is moving up the slope, giving your answer in terms of m and g

The particle is placed on a rough slope which is inclined at an angle of $\theta°$ where $\tan\theta = \frac{4}{3}$ with coefficient of friction μ

It is being pushed up the slope by a horizontal force of Q N.

b) Find the values of μ in terms of Q, m and g such that the particle will not accelerate up the slope.

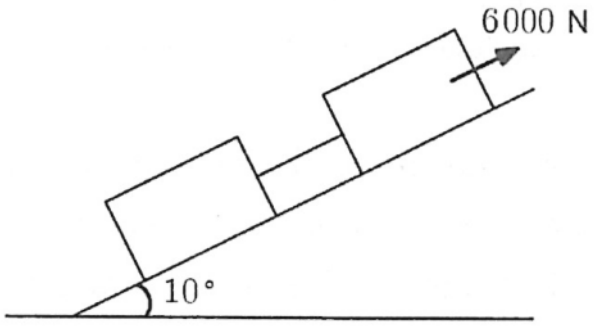
2.55 — Question 20. Particle A of weight W N is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{2}{3}$. The coefficient of friction between the particle and the slope is μ. The block is at the point of sliding down the slope.

a) Show that $\mu = \frac{2}{3}$

Particle A is replaced with particle B which has double the mass of A. The coefficient of friction remains the same.

b) Describe the motion of particle B.

2.55 — Question 21. The following questions use $g = 10$ ms^{-2} A car of mass 2200 kg is towing a trailer of mass 400 kg along a straight flat road which is inclined at 10° to the horizontal. The driving force produced by the engine of the car is 6000 N.

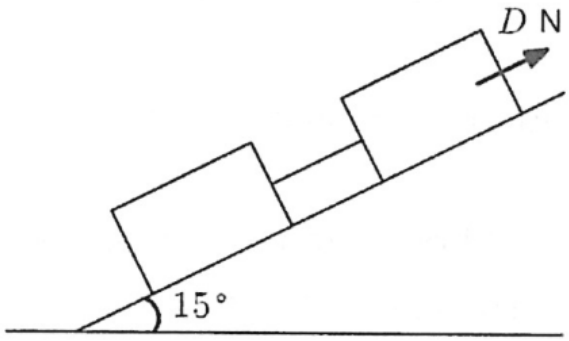


The coefficient of friction between the road and the vehicles is 0.1.

a) Find the deceleration of the car.

b) Find the tension in the tow bar.

2.55 — Question 22. A van of mass 2400 kg is towing a trailer of mass 600 kg along a straight flat road which is inclined at 15° to the horizontal. The van is driving at 8 ms^{-1} and after 5 seconds has reached 12 ms^{-1} due to a driving force of D N.



The coefficient of friction between the road and the vehicles is 0.15.

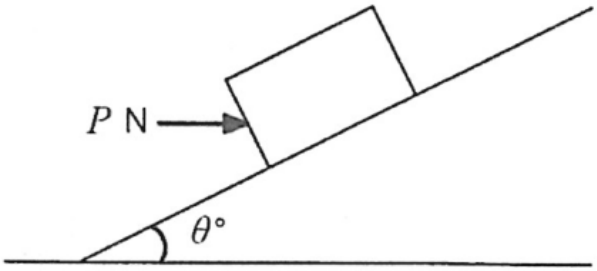
a) Find the value of D

b) Find the tension in the tow bar.

2.55 — Question 23. The following questions use exact values of g A block A has mass m kg and is on a rough slope which is inclined with an angle of 45°. It is being pushed up the slope by a force of P N which is parallel to the slope.
The coefficient of friction between the block and the slope is μ. The block is at the point of sliding.
Show that $\mu = \frac{\sqrt{2}P}{mg} - 1$

2.55 — Question 24. A particle of mass m kg lies on a rough slope which is inclined with an angle of $\theta°$.
It is being pulled upwards by a force of km N which is parallel to the slope, where k is a positive constant.
The coefficient of friction between the particle and the slope is μ
Show that $a = k - \mu g \cos\theta - g \sin\theta$

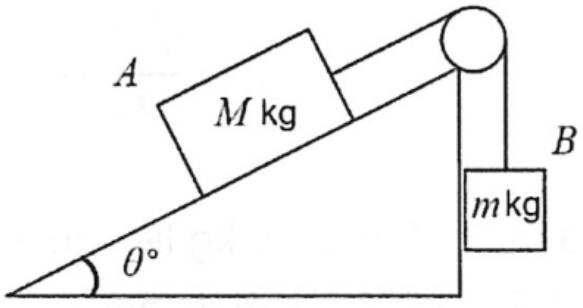
2.55 — Question 25. A particle has weight w N and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$. It is being pushed up the slope by a horizontal force of P N.
The coefficient of friction between the box and the slope is given by μ
The particle is at the point of sliding down the slope.



a) Draw a force diagram.
b) Show that $\mu = \frac{7w-24P}{24w+7P}$

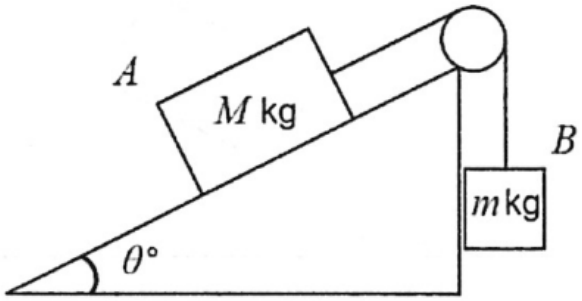
The force P is removed and consequently the block moves down the slope with acceleration $\frac{1}{5}g$ ms^{-2}.
c) Show that $\mu = \frac{1}{12}$

2.56 — Question 1. The following question use $g = 9.8$ ms^{-2} A particle A has mass M kg and is on a smooth slope which is inclined with an angle of $\theta°$. It is being pulled up the slope by a light, inextensible string which is parallel to the slope and passes over a small, smooth, fixed pulley. A second particle B of mass m kg hangs from the pulley freely with the string taut. The system is released from rest.



Find the tension in the string and the acceleration in the system, and state the direction that particle A is moving, given:
a) $M = 1.5, m = 3.9, \theta = 30$
b) $M = 2.1, m = 0.5, \theta = 60$
c) $M = 2, m = 2, \theta = 40$
d) $M = 40, m = 20, \theta = 30$

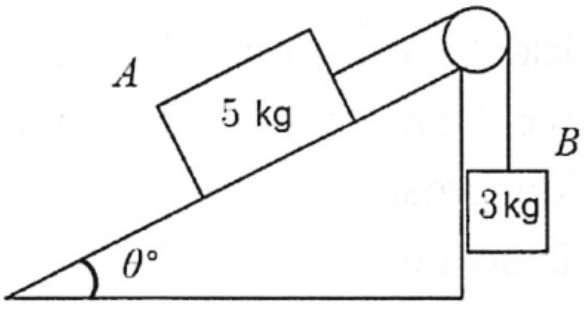
2.56 — Question 2. A particle A has mass M kg and is on a rough slope which is inclined with an angle of $\theta°$. It is being pulled up the slope by a light, inextensible string which is parallel to the slope and passes over a small, smooth, fixed pulley. A second particle B of mass m kg hangs from the pulley freely with the string taut. The coefficient of friction between the slope and the particle is μ. The tension in the string is T N.



Given that the system is released from rest:

a) $M = 2, m = 5, \theta = 30$ and $\mu = 0.85$, find T and the acceleration (and direction) of A
b) $M = 7, \theta = 40, \mu = 0.7, a = 0.4$ (up the slope), find m and T
c) $M = 7, m = 3, \theta = 60, a = 0.1$ (down the slope), find T and μ
d) $M = 2k, m = k$, where k is a positive constant, $\theta = 30, \mu = 0.5$, find a and give the direction of motion.

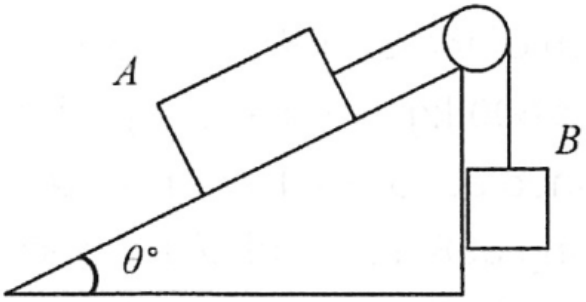
2.56 — Question 3. A box A has mass 5 kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{4}{3}$. A light, inextensible string connects box A to box B which has mass 3 kg. The string is parallel to the slope and passes over a small, smooth, fixed pulley. Box B hangs from the pulley freely with the string taut.



The system is released from rest and the box moves down the slope at 0.8 ms^{-2}.

a) Find the coefficient of friction between the slope and the box.
b) Find the distance travelled by the system in the first 5 seconds, stating any assumptions you have made.
c) Explain what it means for the string to be modelled as light and inextensible.

2.56 — Question 4. A box A has mass m kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$. It is being pulled up the slope by a light, inextensible string which is parallel to the slope and passes over a small, smooth, fixed pulley. A second box B of mass $2m$ kg hangs from the pulley freely with the string taut.



The coefficient of friction between the slope and the box is 0.4. The system is released from rest.

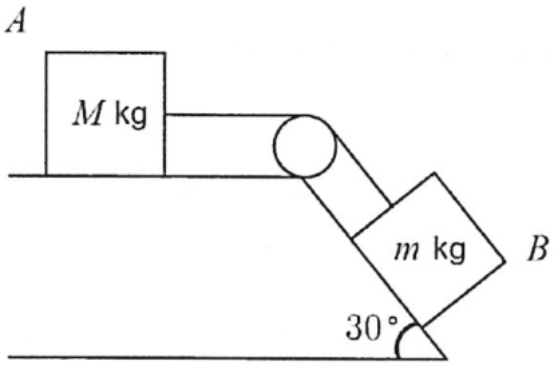
a) Find the tension in the string in terms of m

Given that box B starts 1.5 m above the ground and assuming box A does not reach the pulley. Find

b) the time it takes for box B to reach the ground.

c) the total time box A spends moving from its initial position to when it stops.

2.56 — Question 5. The following questions use $g = 10$ ms^{-2} A block A has mass M kg and is held at rest on a horizontal rough plane. A second block B of mass m kg lies on a smooth inclined plane with an angle of 30°. Blocks A and B are connected by a light, inextensible string which passes over a small, smooth, fixed pulley and is parallel to the surface. The coefficient of friction between block A and the plane is μ. The tension in the string is given by T N.



The system is released from rest. The acceleration of the system is a ms^{-2}. Given

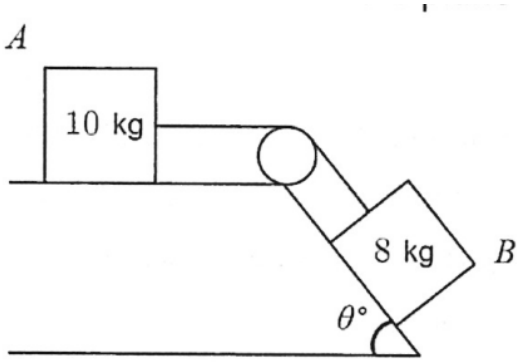
a) $M = 10, m = 3, \mu = 0.2$, find the value of a and T

b) $M = 5, m = 4, a = 1.2$, find the value of T and μ

c) $M = 0.5, m = 2, T = 3$, find the value of a and μ

d) $a = 2, \mu = 0.4, T = 12$ find the value of m and M

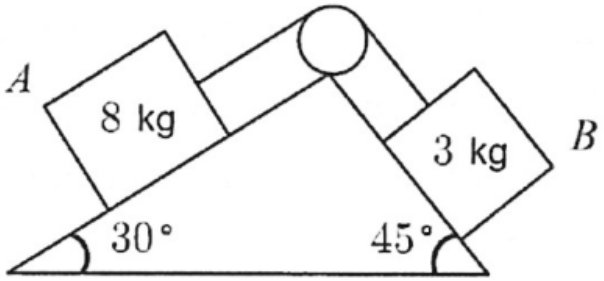
2.56 — Question 6. A block A has mass 10 kg and lies on a horizontal rough plane. A second block B of mass 8 kg lies on a rough inclined plane with an angle of $\theta°$ where $\tan\theta = \frac{4}{3}$. Blocks A and B are connected by a light, inextensible string which passes over a small, smooth, fixed pulley. The coefficient of friction between block A and the plane is 0.2. The coefficient of friction between block B and the plane is 0.4



The system is released from rest and A travels towards the pulley. Find

a) the acceleration of the system.
b) the tension in each of the strings.

2.56 — Question 7. A block A has mass 8 kg and is on a rough slope which is inclined with an angle of 30°. A second block B has mass 3 kg and is on a smooth slope which is inclined with an angle of 45°. Blocks A and B are connected by a light, inextensible string which passes over a small, smooth, fixed pulley.



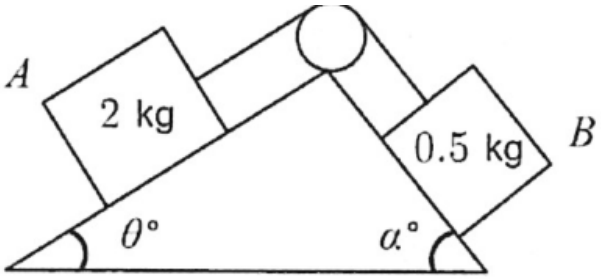
The coefficient of friction between block A and the plane is 0.05. The system is released from rest.

a) Find the acceleration of the system.
b) State the direction in which the blocks are moving.
c) Find the tension in the string.

Block B is replaced by block C with mass m kg and when released it remains in equilibrium.

d) Find the value of m

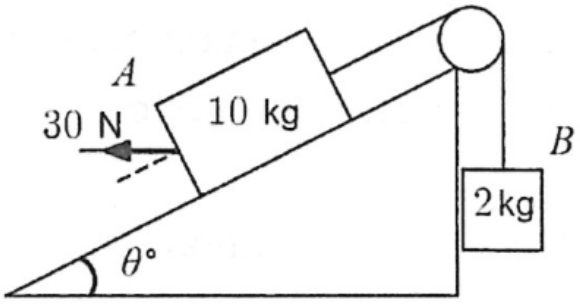
2.56 — Question 8. A block A has mass 2 kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$. A second block B has mass 0.5 kg and is on a rough slope which is inclined with an angle of $\alpha°$ where $\tan\alpha = \frac{4}{3}$. Blocks A and B are connected by a light, inextensible string which passes over a small, smooth, fixed pulley. The rope is parallel to the slopes as shown in the diagram below.



The coefficient of friction between block A and the plane on which it lies is twice the size of the coefficient of friction between block B and the plane on which it lies.
The system is released from rest and A travels with acceleration 0.1 ms^{-2} away from the pulley. Find

a) the coefficient of friction between block A and the plane.
b) the tension in the string.

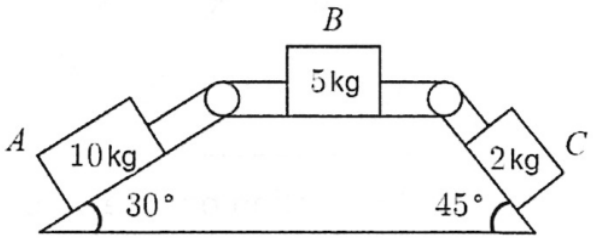
2.56 — Question 9. The following questions use $g = 9.8$ ms^{-2} A box has mass 10 kg is being pulled down a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$ by a light, inextensible, horizontal string which has tension 30 N. The particle is also attached to another light, inextensible string which passes over a small, smooth, fixed pulley to a second box of mass 2 kg which hangs from the pulley freely with the string taut.



The acceleration of the system is 0.4 ms^{-2}.

a) Find the tension in the string connecting the two boxes.
b) Show that the value of the coefficient of friction is approximately 0.519
c) State any modelling assumptions you have used.

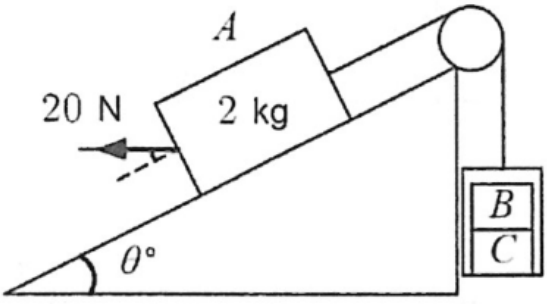
2.56 — Question 10. Box A with mass 10 kg is on a rough plane inclined with an angle of 30° and is connected by a light inextensible string which passes over a small, smooth, fixed pulley to box B. Box B with mass 5 kg is on a rough horizontal plane slope and is connected by a light inextensible string which passes over a small, smooth, fixed pulley to box C. Box C has mass 2 kg lies on a rough plane which is inclined with an angle of 45° to the horizontal.



The coefficient of friction between each of the boxes and the planes are $\mu_A = 0.1, \mu_B = 0.4$ and $\mu_C = 0.2$. The system is released from rest and box A slides down the plane. Find

a) the acceleration of the system
b) the tension in the string between box A and B
c) the tension in the string between box B and C

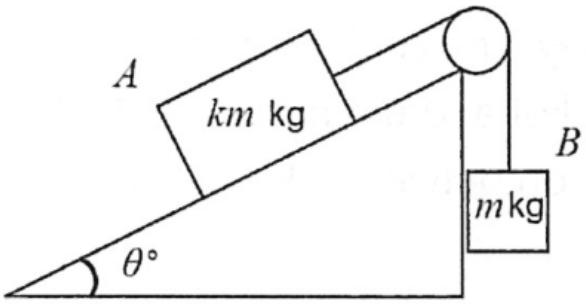
2.56 — Question 11. The following questions use $g = 10$ ms^{-2} A box A has mass 2 kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$. The box is being pulled by a horizontal force of 20 N. Box A is connected to a light, inextensible string which is parallel to the slope and passes over a small, smooth, fixed pulley to a light container. The container has two boxes stacked inside, with box B sitting directly on top of box C. Box B has mass 1 kg and box C has mass 3 kg.



The coefficient of friction between the slope and the box is 0.5

a) Find the acceleration of the system.
b) Find the tension in the string.
c) Find the force exerted on C by B
d) Explain how the system changes if the container is no longer modelled as light.

2.56 — Question 12. The following questions use exact values of g A crate A has mass km kg where k is a positive constant. It lies on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$. It is connected by a rope which is parallel to the slope and passes over a small, smooth, fixed pulley. A second box B of mass m kg hangs with the string taut from the same rope.

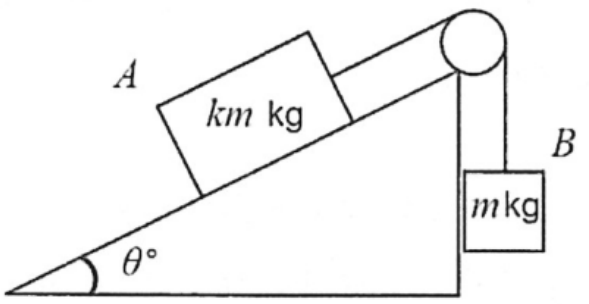


The coefficient of friction between the slope and the box is $\frac{1}{12}$. The system is released from rest and crate A moves down the slope.

a) Show that $a = \left(\frac{4k-13}{13k+13}\right) g$

b) State any modelling assumptions you have made.

2.56 — Question 13. A particle A has mass km kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$. It is being pulled up the slope by a light, inextensible string which passes over a small, smooth, fixed pulley. A second box B of mass m kg hangs from the pulley freely with the string taut. The coefficient of friction between the slope and the box is μ



The system is released from rest and particle A moves 0.5 metres up the slope in 2 seconds.

a) Find the acceleration of particle A.

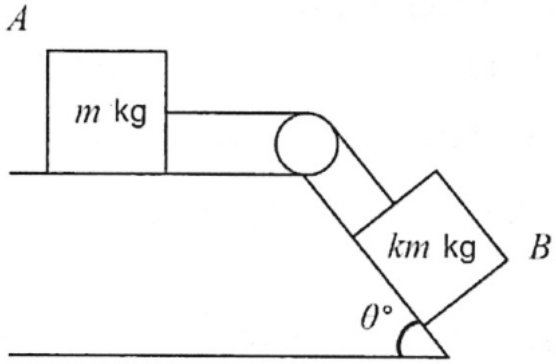
b) Show that the tension in the string is $\left(g - \frac{1}{4}\right) m$

Given that $k = 0.05$

c) Find the value of μ in terms of g

d) State any modelling assumptions you have made.

2.56 — Question 14. A block A has mass M kg and is held at rest on a horizontal rough plane. A second block B of mass m kg lies on a smooth inclined plane with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$. Blocks A and B are connected by a light, inextensible string which passes over a small, smooth, fixed pulley and is parallel to the surface. The coefficient of friction between block A and the plane is μ

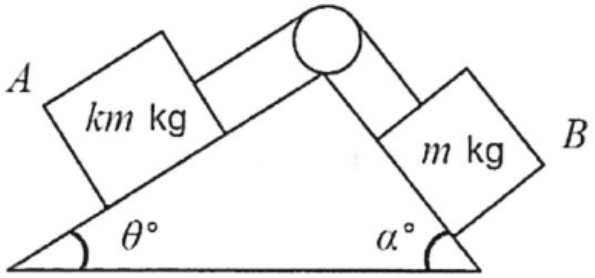


The system is released from rest. The acceleration of the system is a ms^{-2}.

a) Show that $a = \frac{(3k-5\mu)g}{5k+5}$

b) Find the range of values of k for which the system will be moving.

2.56 — Question 15. A block A has mass km kg and is on a rough slope which is inclined with an angle of $\theta°$. A second block B has mass m kg and is on a rough slope which is inclined with an angle of $\alpha°$. Blocks A and B are connected by a light, inextensible string which passes over a small, smooth, fixed pulley.



The coefficient of friction between block B and the plane on which it lies is μ. The coefficient of friction between block A and the plane on which it lies is 3 times larger.

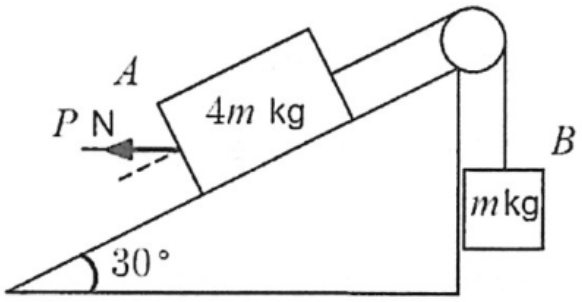
The system is released and is at the point of block B sliding down the slope.

a) Show that $k = \frac{\sin\alpha - \mu\cos\alpha}{\sin\theta + 3\mu\cos\theta}$

Block A is 10% heavier than block B.

b) In the case that $\theta = 30$ and $\alpha = 40$, find the value of μ giving your answer to 3 significant figures.

2.56 — Question 16. A box A has mass $4m$ kg and is being pulled along a rough slope which is inclined with an angle of $30°$ by a force of P N which is horizontal, where $P > 0$
The particle is attached to a light, inextensible string which passes over a small, smooth, fixed pulley to a second box of mass m kg which hangs from the pulley freely with the string taut.



The coefficient of friction between box A and the slope is 0.5. The tension in the string which passes over the pulley is T N.
Find the values of P such that the B is moving upwards.